Course Name: Probability and Statistics 580460

Course Outcome:

CO1: Know the basics of Statistics and apply these techniques in the core course of their study. a
CO2: Understand the fundamental concepts of random variables, and their expectations.
CO3: Understand and apply discrete and continuous probability distributions in real-life situations.
CO4: Apply the Central Limit theorem in real life scenarios.
CO5: Understand and model the relationship between two variables.
CO6: Perform the application of statistical inference in decision-making situations.

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End Term Examination, Even Semester 2024-25
B. Tech. (CS), II – Year, IV - Semester
BMAS 0108 : Probability and Statistics

Time: 3 Hours **Maximum Marks: 50**

Note – 1. Attempt ALL the Sections.

Section – A

*Attempt **All** Questions* 4×5=20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Find the median from the following table: Marks: 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 No. of students: 2 | 18 | 30 | 45 | 35 | 20 | 6 | 3	4	1	A	P
2	Suppose that the dataset consists of these hypothetical test scores obtained in an examination: **29, 15, 90, 79, 85, 75, 91, 93, 98, 93.** Construct a box-plot for the above data and check for outliers if any.	4	1	U	C
3	The following is the distribution function of a discrete random variable: X: -3 | -1 | 0 | 1 | 2 | 3 | 5 | 8 $F(X)$: 0.10 | 0.30 | 0.45 | 0.65 | 0.75 | 0.90 | 0.95 | 1.00 (i) Find the probability distribution of X. (ii) Find $P(1 \leq X \leq 8)$.	4	2	An	F

4	Find the Karl Pearson's coefficient of correlation between the values of x and y given below: x: 1, 3, 5, 7, 8, 10 y: 8, 12, 15, 17, 18, 20	4	5	E	P
5	The following data is collected on two characters: Those who go to Cinema / Those who don't go to Cinema Literate: **230** / **148** Illiterate: **151** / **471** Based on this, can you conclude that there is no relation between the habit of going to cinema and literacy? [Given: **tabulated** $\chi^2_{0.05} = 3.841$ for **1 d.f.**] **OR** A random sample of 16 values from a normal population showed a mean of 41.5 inches and the sum of squares of deviations from this mean equal to 135 square inches. Show that the assumption of a mean of 43.5 inches for the population is not reasonable. [Given: tabulated $t_{0.05} = 2.131$ for **15 d.f.**].	4	6	An	C

Section – B

*Attempt **All** Questions* 3×5=15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	If mean and variance of a binomial distribution are 4 and 2 respectively, find the probability of (i) exactly 2 success (ii) less than 2 success	3	3	A	P
7	If the probability that a target is destroyed on any one shot is 0.5, what is the probability that it would be destroyed on 6th attempt?	3	3	E	C
8	State the conditions under which the Poisson distribution becomes a limiting distribution of Binomial distribution.	3	6	A	M
9	Explain Type I and Type II error briefly.	3	4	U	C

10	Two lines of regression are given by $x + 2y = 5$ *and* $2x + 3y = 8$, calculate the coefficient of correlation between x and y.	3	5	An	C

Section – C

*Attempt **All** Questions* 5×3=15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	In a certain factory turning out razor blades, there is a small chance of 0.002 for any blade to be defective. The blades are supplied in packets of 20, use Poisson distribution to calculate the approximate number of packets containing no defective blade, one defective blade and two defective blades respectively in a consignment of 10,000 packets.	5	3	U	P
12	A sample of 100 dry battery cells tested to find the length of life produced the following results: mean = 12 hours, $\sigma = 3$ hours. Assuming the data to be normally distributed, what percentage of battery cells are expected to have life: **(i).** more than 15 hours. **(ii).** less than 6 hours. **(iii).** between 10 *and 14 hours?* [Given: $\mathbf{P(0 \le Z \le 1) = 0.3413}$, $\mathbf{P(0 \le Z \le 2) = 0.4772}$ **, and** $\mathbf{P(0 \le Z \le 0.67) = 0.2485}$. where Z is a standard normal variate].	5	3	A	P
13	The following table gives age (x) in years of cars and annual maintenance cost (y) in hundred rupees. Estimate the maintenance cost for a 4 year old car after finding the regression equation.	5	5	E	P

x	1	3	5	7	9
y	15	18	21	23	22